Supplementary Material for *Timestep-Conditioned Adapter Scaling for Multi-view Diffusion Texture Generation*

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This supplement supports the revised 01549 manuscript. Fresh C is an image-space comparison on 300 UIDs disjoint from the original 01549 evaluation/probe IDs, the 1,118-object training list, and clean-v2 evaluation IDs. Its primary conditions and same-cohort addendum share one frozen adapter checkpoint, runner source, seed policy, inputs, target views, and metric implementation; the addendum is not an independent replication. Fresh C uses a controlled-retraining checkpoint and is not presented as a numerical reproduction of historical Table 4. Historical cohorts and interface checks are reported separately. Object is the inferential unit; multiple views and wrapper records do not increase N.

S1. Figure asset credits and selection status

Main Figures 4–5 use Fresh C reference and method-level RGB outputs in a compact context-plus-shared-ROI layout. The original six-view, seven-condition panels, including No Adapter, LLH, and Generic Linear, remain in Supplementary Figure S1. The selected examples are outcome-ranked illustrations, not a representative sample. Image, checkpoint, run, seed, ROI, and output hashes are listed in `FRESH_C_FIGURE_SOURCE_ROI_MANIFEST.csv`.

The Fresh C No Adapter condition sets the geometry-adapter scale to zero. In the MVPainter wrapper, when `_adapter_scale` is unset the geometry residual is applied without a scale multiplier. These are different settings: No Adapter is not a rendered comparison against the unmodified native-default adapter path. Fresh C contains no separate output for that native-default path, so this part of the R1.1 request remains partially unresolved.

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Table S1: Attribution for selected Fresh C assets in main Figures 4–5 and Supplementary Figure S1.

Use	Full attribution
Figure 4(a), C3–GFL endpoint-favourable example	*Building 03* by Shalmon, via Sketchfab (`https://sketchfab.com/3d-models/c76ac44df995482185077da81939b306`), licensed CC BY 4.0 (`https://creativecommons.org/licenses/by/4.0/`). Fresh C target-view RGB outputs are cropped to the same ROI and arranged without color or detail adjustment.
Figure 5(a), C3-versus-fixed-scale example	*Octaèdre* by Segolene Delamare, via Sketchfab (`https://sketchfab.com/3d-models/0571e6db30c84b1593bc55a8e145a430`), licensed CC BY 4.0 (`https://creativecommons.org/licenses/by/4.0/`). Fresh C target-view RGB outputs are cropped to the same ROI and arranged without color or detail adjustment.

Use	Full attribution
Figure 5(b), Generic Linear counterexample	*Cube 01* by b0rF4x, via Sketchfab (`https://sketchfab.com/3d-models/06fa4974f8cd4d4ea792e98fa1521457`), licensed CC BY 4.0 (`https://creativecommons.org/licenses/by/4.0/`). Fresh C target-view RGB outputs are cropped to the same ROI and arranged without color or detail adjustment.
Figure 4(b) and Figure S1(a), adverse C3–GFL example	*Boat Low Poly Game Ready* by INinja, via Sketchfab (`https://sketchfab.com/3d-models/d0e116a62bfc4cdd9edf60671073b306`), licensed CC BY 4.0 (`https://creativecommons.org/licenses/by/4.0/`). Fresh C target-view RGB outputs are cropped to the same ROI and arranged without color or detail adjustment.
Figure S1(b), color/detail mismatch example	*E5-5-2* by artsiyu, via Sketchfab (`https://sketchfab.com/3d-models/f6979d7ff8e54c4daf00fee46f09a139`), licensed CC BY 4.0 (`https://creativecommons.org/licenses/by/4.0/`). The complete Fresh C contact sheet is retained for a broader-condition comparison.

S2. Complete Fresh C paired image-space statistics

Table S2 reports the complete seven-endpoint paired results for C3–GFL and each of the three named addendum contrasts. The primary C3–GFL FG-PSNR result is unadjusted as the single registered primary endpoint. The six registered secondary C3–GFL endpoints use their Holm family. The addendum's FG-PSNR and FG-LPIPS contrasts use the frozen six-test primary family; the other five endpoints per contrast use the separate 15-test secondary Core-7 family. All contrasts are mean A-minus-B differences with 10,000 object bootstrap resamples and pointwise two-sided percentile 95% CIs. Positive values favour A for PSNR/SSIM; negative values favour A for LPIPS. Adjusted p-values are from the frozen analysis authority. "A favourable" is the fraction of object-level differences in A's favourable direction and is descriptive.

Table S2: Complete paired object-level image-space results for the primary C3–GFL contrast and the three named same-cohort addendum contrasts. Differences are A minus B; favorable-object percentages respect endpoint direction.

Endpoint	Mean difference [95% CI]	Median	A favorable	Adjusted p
C3 minus GFL				
FG-PSNR (dB)	+0.522 [+0.404, +0.641]	+0.555	68.7%	primary
FG-LPIPS	-0.005345 [-0.006326, -0.004403]	-0.004547	71.0%	3.2×10^{-22}
FG-SSIM	+0.02453 [+0.02120, +0.02805]	+0.02004	84.3%	5.9×10^{-33}
Edge-SSIM	+0.004494 [+0.003411, +0.005586]	+0.003613	69.0%	9.4×10^{-14}
Full-view PSNR (dB)	+0.271 [+0.170, +0.373]	+0.220	59.3%	4×10^{-07}
Full-view LPIPS	-0.003880 [-0.004926, -0.002873]	-0.003029	64.3%	8×10^{-13}
Full-view SSIM	+0.005619 [+0.004627, +0.006704]	+0.005205	83.0%	7.9×10^{-22}
C3 minus Generic Linear				
FG-PSNR (dB)	-0.593 [-0.837, -0.351]	+0.074	51.7%	7.2×10^{-06}
FG-LPIPS	+0.013512 [+0.011601, +0.015487]	+0.012229	25.7%	1.3×10^{-32}
FG-SSIM	-0.02810 [-0.03197, -0.02439]	-0.02521	20.0%	2.9×10^{-35}

Endpoint	Mean difference [95% CI]	Median	A favorable	Adjusted p
Edge-SSIM	-0.010693 [-0.013004, -0.008495]	-0.008967	33.7%	1.6×10^{-17}
Full-view PSNR (dB)	-0.503 [-0.686, -0.322]	-0.139	46.7%	2.7×10^{-07}
Full-view LPIPS	-0.000897 [-0.002578, +0.000862]	-0.001307	54.3%	0.611
Full-view SSIM	+0.000046 [-0.000984, +0.001011]	+0.001213	57.0%	0.927
LLH minus C3				
FG-PSNR (dB)	+0.621 [+0.391, +0.855]	-0.153	46.7%	1.2×10^{-06}
FG-LPIPS	-0.012140 [-0.013861, -0.010480]	-0.010837	75.0%	9×10^{-34}
FG-SSIM	+0.01658 [+0.01311, +0.02003]	+0.01307	67.0%	9×10^{-18}
Edge-SSIM	+0.009033 [+0.006867, +0.011296]	+0.006741	63.3%	2.3×10^{-13}
Full-view PSNR (dB)	+0.816 [+0.639, +0.998]	+0.419	59.3%	3.2×10^{-16}
Full-view LPIPS	+0.004909 [+0.003266, +0.006510]	+0.004538	36.7%	3.5×10^{-08}
Full-view SSIM	-0.004330 [-0.005176, -0.003482]	-0.004563	27.7%	5.9×10^{-20}
LLH minus Generic Linear				
FG-PSNR (dB)	+0.028 [-0.015, +0.070]	+0.017	52.3%	0.201
FG-LPIPS	+0.001372 [+0.000762, +0.002081]	+0.000176	46.7%	0.000114
FG-SSIM	-0.01152 [-0.01337, -0.00996]	-0.00945	8.0%	2.6×10^{-31}
Edge-SSIM	-0.001660 [-0.002096, -0.001256]	-0.001648	27.3%	1.2×10^{-12}
Full-view PSNR (dB)	+0.313 [+0.278, +0.347]	+0.304	87.3%	2×10^{-47}
Full-view LPIPS	+0.004011 [+0.003390, +0.004684]	+0.003904	18.0%	1.7×10^{-26}
Full-view SSIM	-0.004284 [-0.005217, -0.003543]	-0.003432	5.7%	8.5×10^{-20}

S3. Supporting layer/window and schedule evidence

A2 is exploratory discovery evidence; A3 is a higher-dose stress regime on the same underlying cohort. A3b is a bounded Fresh B layer/window analysis. Object-cluster tests detect response heterogeneity under the tested interventions, but they do not identify a dose-independent layer-by-time mechanism or an optimal schedule.

Table S3: Object-cluster Wald interaction summaries for the tested layer-by-window maps. Cell RMSE and interaction share describe these observed maps; they are not causal effect sizes.

Map	Endpoint	Wald statistic	Interaction share; cell RMSE	Wald p (A3b Holm)
A2	FG-LPIPS	481.081	43.5%; 0.001695	8.04×10^{-99}
A2	FG-PSNR	1879.963	56.8%; 0.1222 dB	$< 10^{-300}$
A3	FG-LPIPS	657.237	11.9%; 0.00643	1.14×10^{-136}
A3	FG-PSNR	1389.118	0.8%; 0.207 dB	1.28×10^{-294}
A3b	FG-LPIPS	225.015	14.0%; 0.000342	5.37×10^{-43}
A3b	FG-PSNR	525.700	36.0%; 0.0205 dB	3.43×10^{-107}

For A3b, the three layer groups had no common support across 2,250 layer-step rows. Their central 90% realized-dose ranges were deep 56,359–79,927, middle 11,421–19,336, and shallow 2,816–5,193. Dose-

adjusted extrapolations do not supply the missing common support. These results therefore support only measurable response heterogeneity under the tested intervention regime.

The separate Fresh B cohort ($N = 150$) supplies schedule-location context for LLH, not evidence for C3. Table S4 preserves both registered LLH comparisons and the post-outcome HLL–LLL diagnostic. Paired intervals use 10,000 object-bootstrap resamples (seed 20261002). The LLH–LFM-EXACT intervals lie within the protocol's practical margins (FG-PSNR ±0.5 dB; FG-LPIPS ±0.01); this is a budget-explained interpretation within those margins, not exact equality. HLL–LLL is exploratory and its endpoint directions differ.

The Fresh B protocol-lock file states a freeze time of 2026-10-05 08:40 UTC, while the earliest preserved commit containing that file is dated 2026-10-06 00:52 UTC; its parent does not contain the file. This history does not establish when the lock was first created outside Git, but it cannot authenticate pre-run availability. Accordingly, Fresh B temporal contrasts, the A3b map, and the post-lock generic extension are treated as retrospective sensitivity evidence in this paper, even where their calculations follow the recorded comparison family. They are not described as verified preregistered confirmation.

Table S4: Fresh B paired object-level LLH comparisons, classified here as retrospective sensitivity because pre-run lock availability cannot be authenticated from the preserved history. Effects are A minus B; favorable rates respect endpoint direction. Holm values reproduce the recorded family.

Contrast	Endpoint	Mean [95% CI]	A favors	p / status
LLH–LFM-EXACT	FG-PSNR	+0.325 [+0.209,+0.446] dB	54.0%	raw ≤ 0.0001
LLH–LFM-EXACT	FG-LPIPS	−0.00460 [−0.00563, −0.00357]	76.0%	raw ≤ 0.0001
LLH–HLL	FG-PSNR	+0.687 [+0.404,+0.981] dB	45.3%	≤ 0.0004; median −0.104 dB
LLH–HLL	FG-LPIPS	−0.01098 [−0.01349, −0.00851]	80.7%	≤ 0.0004
LLH–LLL	FG-PSNR	+0.734 [+0.597,+0.875] dB	79.3%	≤ 0.0004
LLH–LLL	FG-LPIPS	−0.00845 [−0.01029, −0.00659]	82.0%	≤ 0.0004
HLL–LLL (exploratory)	FG-PSNR	+0.047 [−0.120, +0.208] dB	66.0%	0.5696; median +0.476 dB
HLL–LLL (exploratory)	FG-LPIPS	+0.00253 [+0.00155,+0.00354]	40.0%	≤ 0.0002; median +0.00139

The post-lock generic-schedule extension on this same Fresh B cohort is a retrospective sensitivity analysis, not independent confirmation. Table S5 reports the recorded budget-matched comparisons and endpoint-matched linear counter-result. Nominal requested-scale means do not equalize post-scale residual dose; these LLH comparisons do not establish dose-adjusted schedule superiority or strengthen C3 specifically.

Table S5: Post-lock Fresh B generic-schedule extension (same $N = 150$ cohort). Effects are LLH minus comparator. Budget-matched schedules form the locked eight-schedule family; endpoint-matched linear is the registered counter-comparison.

Contrast	Endpoint	Mean [95% CI]	A favors	Holm p
LLH–budget linear	FG-PSNR	+0.0967 [+0.0594,+0.1357] dB	65.3%	Holm $p \leq 0.0008$
LLH–budget linear	FG-LPIPS	-0.0009 [-0.0013,-0.0004]	65.3%	Holm $p = 0.0008$
LLH–budget cosine	FG-PSNR	+0.4629 [+0.3526,+0.5780] dB	74.7%	Holm $p \leq 0.0008$
LLH–budget cosine	FG-LPIPS	-0.0059 [-0.0071,-0.0046]	80.0%	Holm $p \leq 0.0008$
LLH–budget trapezoid	FG-PSNR	+0.4404 [+0.3334,+0.5521] dB	74.0%	Holm $p \leq 0.0008$
LLH–budget trapezoid	FG-LPIPS	-0.0057 [-0.0070,-0.0045]	78.0%	Holm $p \leq 0.0008$
LLH–budget Gaussian	FG-PSNR	+0.4711 [+0.3584,+0.5892] dB	74.7%	Holm $p \leq 0.0008$
LLH–budget Gaussian	FG-LPIPS	-0.0058 [-0.0071,-0.0045]	79.3%	Holm $p \leq 0.0008$
LLH–endpoint linear	FG-PSNR	-0.0373 [-0.0829,+0.0086] dB	–	Holm $p = 0.1150$

Contrast	Endpoint	Mean [95% CI]	A favors	Holm p
LLH–endpoint linear	FG-LPIPS	+0.00109 [+0.0004,+0.0019]	–	Holm $p = 0.0024$

S4. Requested/applied scale and residual-correction accounting

Table S6 lists representative Fresh C C3 records at five denoising steps. The native caps are 3.0 (deep), 3.5 (middle), and 0.8 (shallow). A cap-active fraction of 100% means all logged records in that group/step were clipped. The applied correction norm is computed after applying the effective scale and depends on the residual itself; it is a logged implementation quantity, not an independent dose or quality outcome. The 750-row source accounting table contains all 50 steps, three groups, and five strategies.

Table S6: Representative Fresh C requested/applied scale and post-scale residual-correction accounting. Median correction norms are descriptive log summaries over implementation records; wrappers, objects, and timesteps are not independent experimental units.

Step	Layer group	Requested	Applied	Cap active	Median correction L_2
0	Deep	1.25	1.25	0%	1309.19
0	Middle	1.25	1.25	0%	468.97
0	Shallow	1.25	0.80	100%	30.86
12	Deep	1.25	1.25	0%	1238.42
12	Middle	1.25	1.25	0%	661.91
12	Shallow	1.25	0.80	100%	42.79
24	Deep	2.50	2.50	0%	2006.94
24	Middle	2.50	2.50	0%	1551.26
24	Shallow	2.50	0.80	100%	52.51
37	Deep	1.25	1.25	0%	672.23
37	Middle	1.25	1.25	0%	631.83
37	Shallow	1.25	0.80	100%	55.62
49	Deep	1.25	1.25	0%	659.98
49	Middle	1.25	1.25	0%	197.82
49	Shallow	1.25	0.80	100%	34.72

S5. Historical and cross-interface boundaries

The completed `MAIN-STAGE-276` follow-up contains 1,104 records (276 objects by four schedules), with 50 Euler steps, seed 42, unique-six target views, shared initial latents, and a common runner within the run. Its final run manifest records completion with exit code 0. The follow-up was specified after inspection of earlier clean-v2 results and was not blind. It is reported separately from Fresh C and is not an independent replication of the original 01549 development probe.

Here, temporal L–L–H means that the requested values are low in early and middle steps and high in late steps. This is distinct from the layer-conditioned Fresh C LLH addendum condition described in the main paper. With 50 steps, the code assigns 17/16/17 steps to early/middle/late. C3 therefore has deep/middle applied-scale sums of 82.5; temporal L–L–H has 83.75 because its high stage occupies 17 rather than 16 steps; the shallow applied sum is 40 for both after the 0.8 cap. The fixed-mean comparator has a deep/middle sum of 83.33 and a shallow sum of 40. These are near-matched scalar sums, not exactly equal sums or matched residual-norm doses.

Table S7 gives all six formal paired endpoints for C3 versus temporal L–L–H and C3 versus the fixed-mean comparator. Effects are C3 minus comparator; positive values favor C3 for PSNR/SSIM, and negative values favor C3 for LPIPS. Temporal L–L–H outperforms C3 on all six formal endpoints in this post-inspection stage-placement experiment: it has higher PSNR/SSIM and lower LPIPS endpoints. In contrast, C3 exceeds fixed-mean on five of six formal endpoints. These results show that the measured response varies with tested stage placement; they do not establish a universal schedule winner. The contrasts do not isolate placement from the small applied-scale-sum differences or the resulting residual-state trajectory, and post-scale residual norms are not matched.

Table S7: Formal paired comparisons from the strict-276 stage-placement follow-up. Effects are C3 minus comparator; 95% percentile-bootstrap intervals are object-level.

Contrast	Endpoint	Mean difference [95% CI]
C3 minus temporal L–L–H		
	Full-view PSNR (dB)	-0.268100 $[-0.311101, -0.227797]$
	FG-PSNR (dB)	-0.177084 $[-0.230539, -0.130784]$
	FG-SSIM	-0.008851 $[-0.011455, -0.006291]$
	Edge-SSIM	-0.035301 $[-0.036581, -0.034031]$
	FG-LPIPS	$+0.001896$ $[+0.001205, +0.002578]$
	Full-view LPIPS	$+0.008658$ $[+0.007465, +0.009831]$
C3 minus fixed-mean		
	Full-view PSNR (dB)	$+0.556369$ $[+0.525520, +0.589220]$
	FG-PSNR (dB)	$+0.502875$ $[+0.462018, +0.542244]$
	FG-SSIM	$+0.032083$ $[+0.028966, +0.035172]$
	Edge-SSIM	$+0.001696$ $[+0.000611, +0.002782]$
	FG-LPIPS	-0.003797 $[-0.004341, -0.003260]$
	Full-view LPIPS	$+0.001420$ $[+0.000778, +0.002072]$

The saved-artifact Full-SSIM branch is kept separate from the formal pre-save metrics above: C3 minus temporal L–L–H is -0.001697 (95% CI $[-0.002047, -0.001340]$); C3 minus fixed-mean is $+0.007486$ ($[+0.006736, +0.008262]$). These saved-artifact values are not pooled with the formal metric column.

A retrospective strict-276 Core-7 summary uses the clean-v2 object list (SHA-256 `a6aa8ab6e475763e1b5e67dc8c712ec8f1940e3952d65897c820887554bbd044`) and the same recorded checkpoint SHA as Fresh C. It shares the 276-object cohort/checkpoint lineage with the separate `MAIN-STAGE-276` profile-placement follow-up, so those results are not independent cohort replications. The row audit verifies 276 unique object indices per condition and identical object labels by index. However, Core-7 does not establish one common runner or fully shared inputs across all seven conditions: No Adapter and GFH are the two newly rerun arms with per-object input-tensor hashes; their recorded runner hash is `ae5d5664e7d0ebcd4bd84c7c7d688c4a8962c6d74463fe88e7a866404911e7e0`. C3 records a different runner hash, `d4af0c1d7f42168a585b505a6d775ea3ff7ffb18345156ebf358f272025fa7f2`; GFL and the three layer conditions do not establish the same runner hash, and their serialized rows do not contain per-object input-tensor hashes. Thus Core-7 is a retrospective diagnostic, not a same-runner confirmatory comparison. It is not pooled with Fresh C, and the shared checkpoint hash does not make it a numerical reproduction of the original 01549 Table 4 result.

Table S8 gives the seven-condition means for the requested image-space endpoints. These stored metrics show an endpoint trade-off: C3 has a higher Full-view PSNR mean than GFH but a lower FG-PSNR mean. The Core-7 Layer LLH row is distinct from Fresh C's layer-conditioned LLH addendum and Fresh B's temporal LLH condition. Table S9 reports selected object-indexed contrasts. C3–GFH intervals were recalculated from the serialized per-object rows after inspection of the Core-7 summary; they are post-hoc, nominal pointwise intervals. The other listed contrasts reproduce the separate Core-7 paired-statistics file. The intervals describe the saved metric rows and do not overcome the common-runner/input-hash limitations.

Table S8: Retrospective Core-7 strict-276 condition means. Rows contain 276 object-level metrics per condition. The collection has incomplete common runner/input identity as described above; means are descriptive and are not Fresh C or an independently verified same-runner comparison.

Condition	Full-view PSNR↑	FG-PSNR↑	FG-LPIPS↓	FG-SSIM↑	Edge-SSIM↑
No Adapter	10.563	8.826	0.2059	0.4799	0.4794
GFL	18.229	10.731	0.1917	0.3987	0.5201
GFH	18.242	13.153	0.1677	0.4783	0.5538
C3	18.968	11.939	0.1817	0.4296	0.5278
Layer Fixed Mean	19.516	13.100	0.1698	0.4550	0.5363
Layer LHL	19.635	12.974	0.1719	0.4445	0.5273
Layer LLH	20.828	13.975	0.1610	0.4620	0.5476

Table S9: Selected retrospective Core-7 paired contrasts (first condition minus second). Intervals are pointwise 95% object-bootstrap percentile intervals with 10,000 resamples and seed 20260930; they are unadjusted and descriptive. The C3–GFH intervals were recalculated post hoc. The final column counts objects where the first condition is favorable for PSNR.

Contrast	Endpoint	Mean difference [95% CI] (dB)	Objects favoring first
C3–GFH	Full-view PSNR	$+0.725229$ $[+0.647134, +0.806467]$	240/276
C3–GFH	FG-PSNR	-1.214235 $[-1.313952, -1.114705]$	25/276
C3–GFL	FG-PSNR	$+1.207771$ $[+1.116967, +1.294989]$	259/276
LLH–GFH	FG-PSNR	$+0.822292$ $[+0.616795, +1.033035]$	148/276
LLH–C3	FG-PSNR	$+2.036527$ $[+1.806804, +2.260756]$	227/276

The frozen MV-Adapter probe analyzed 98 of 99 planned objects; the omitted input was a frozen GLB with path geometry and no triangle faces, so no substitute was used. The production additive-residual bootstrap interaction statistic is retained in the source audit but its null distribution is centered at or above the observed statistic; this test is invalid for interaction inference. The post hoc exploratory object-cluster robust Wald reanalysis uses 8 degrees of freedom and gives FG-LPIPS $W = 13.322$, raw $p = 0.101$, Holm $p = 0.303$; PSNR $W = 7.995$, raw $p = 0.434$, Holm $p = 0.868$; FG-SSIM $W = 16.243$, raw $p = 0.039$, Holm $p = 0.156$; and CIEDE2000 $W = 3.945$, raw and Holm $p = 0.862$. No metric survives the exploratory four-metric correction. Small adverse shallow FG-LPIPS changes occur in late windows W3–W5 within the registered 15-cell family; the tested scale range may be too weak to reveal larger responses. This bounded, exploratory result does not establish absence of interaction or equivalence. MVDiffusion was evaluated separately on 75 objects using a different correspondence-aware conditioning interface; its within-interface results were mixed/negative across structure, color, and texture endpoints. Neither interface is a matched architecture comparison, so these analyses do not establish architecture causality or cross-backbone generalization. Each system uses its own conditioning and runner, and the measurements are reported within interface rather than as a controlled shared-checkpoint comparison.

The first-round 01549 Section 4.7 adapter-dependence analysis is retained as a separate within-family result. It reported three outcomes for uniform high scale across adapter regimes: texture flattening without per-layer caps, high-frequency artifact amplification under a strong-residual adapter with per-layer caps, and negligible effect for a weakened adapter. Its same-checkpoint cap ablation reported both flattening and amplification, and its residual-magnitude sweep (0.1×–3× output projection) reported increasing intervention strength with residual magnitude. A six-object stage ablation reported that the most sensitive stage differed between the main and artifact-prone adapters. These first-round results were explicitly described as adapter-dependent; they are retained as historical, within-architecture evidence and do not answer the current cross-backbone question.

A separate 76-object MV-Adapter study tested layer-wise L–LLH under an alternative budget-neutral contiguous partition B′ (per-point multipliers $[0.52018, 0.52018, 1.47982, 1.47982]$, weighted mean 1).

Relative to the corresponding global controls, the B′ results are mixed across endpoints: the layer condition favors the global controls on FG-SSIM, favors the layer condition on the benefit-oriented CIEDE2000 and GT-relative texture summaries, and does not separate on FG-PSNR, Edge-SSIM, or FG-LPIPS. Its difference from the original partition is not resolved on any reported metric. These data describe this MV-Adapter layer intervention only and are not evidence for the TCAS timestep profile or for a general mapping rule.

The companion M3 profile uses $[0.41953, 0.41953, 1.19349, 1.19349]$, whose mean multiplier is approximately 0.8065. It changes both group assignment and total nominal residual budget, so it is not a pure mapping comparison with B′. Its audit classified 16 of 18 direction summaries as stable relative to the original partition, with one small Edge-SSIM attenuation and one near-zero FG-LPIPS sign change. Because the preserved protocol and result rows first appear in the same commit, M3 is retained as retrospective sensitivity. Neither M3 nor B′ establishes mapping robustness or cross-backbone generalization.

S6. Historical N=20 unseen-view results

Table S10 reports the bounded historical N=20 GLB results. The subset comprises 20 UV-compatible objects from the original 24-object pool; four inputs were excluded for lack of UVs. Each object is the paired unit after averaging 11 frozen unseen views. Rendering uses unlit base color with native sampler interpretation. Endpoint directions are mixed; these statistics do not establish population-level 3D superiority, PBR/material fidelity, seam quality, or human preference. The source GLB assets and internal montage are not included in this supplement.

Table S10: Historical N=20 unlit base-color paired unseen-view summaries (C3 minus comparator). The 20-object UV-compatible subset comes from the historical 24-object pool; four objects were excluded for missing UVs. Each object is averaged over 11 frozen unseen views. Intervals are 10,000-object-bootstrap percentile CIs (seed 20261005), descriptive and unadjusted.

Comparator	Endpoint	Mean difference [95% CI]	Objects favoring C3
GFL	FG-PSNR (dB)	-0.066 [-0.413, +0.290]	5/20 (25%)
GFL	FG-LPIPS	-0.004719 [-0.007504, -0.002544]	17/20 (85%)
GFL	CIEDE2000	+0.133 [-1.161, +1.319]	6/20 (30%)
GFH	FG-PSNR (dB)	+0.084 [-0.290, +0.426]	14/20 (70%)
GFH	FG-LPIPS	+0.007709 [+0.004230, +0.012101]	4/20 (20%)
GFH	CIEDE2000	-1.027 [-2.224, +0.229]	14/20 (70%)
No Adapter	FG-PSNR (dB)	+3.135 [+1.721, +4.493]	18/20 (90%)
No Adapter	FG-LPIPS	-0.014519 [-0.029860, -0.001326]	16/20 (80%)
No Adapter	CIEDE2000	-12.508 [-18.556, -6.529]	18/20 (90%)

S7. Selected negative visual evidence and unresolved 3D checks

Figure S1 presents selected illustrative Fresh C failure cases. Panel (a) is an object-level case where GFL favours C3 on the paired image-space comparison. Panel (b) illustrates a color/detail mismatch and repeated pattern in the No Adapter output; it is not a C3-specific failure claim. Neither panel estimates failure prevalence. Full attributions appear in Table S1.

The attempted Fresh C 24-object GLB bake did not pass its input/output gate: 9/24 sources had out-of-range native UVs, and 5 assets / 27 primitives used unsupported BLEND. No valid Fresh C 24-object bake results were generated. The seam endpoint is retired because the required seam truth set and native full-panel validation did not pass. No full PBR comparison is available. These checks are limitations, not evidence of method failure or success.

(a) Selected C3–GFL adverse example



(b) Selected color/detail mismatch example



Figure S1: Selected illustrative failure cases from the rights-cleared Fresh C 2D pool. They are not representative estimates and do not support PBR, seam, or human-preference claims.

S8. Statistical authority and reproduction map

Each numerical result in this Supplement is mapped to the internal claim ledger and scoped result sources. Fresh C primary contrasts use object-level deltas and a locked analysis; requested/applied-scale records use the per-step accounting table. Layer/window and Fresh B results are maintained in separate source summaries, with Fresh B classified as retrospective sensitivity because its lock chronology cannot be authenticated from preserved Git history. The texture-complexity cohort and the color-diagnostic set are separate from Fresh C and from each other. The historical N=20 GLB-native summary has a separate analysis record; no N=20 image assets are redistributed. Original Table 9 and Table 3 below are reproduced only as historical source text with explicit provenance limits; neither is treated as a newly verified current result.

Fresh C identity details. The 300-object Fresh C list is disjoint by UID from the original eval300, including the 24-object development probe, the 1,118-object training list, and the clean-v2 eval300. The primary four-condition run and same-cohort Generic Linear/LLH addendum use the same frozen checkpoint SHA-256 `0618d6b284ab47aa16d7a89dc447f5ba4455ff6c9d918e5e129478a99e0114c0`, runner source SHA-256 `e5c28e915ba137d3541eb4c032daaf62a798332d63b32917b749fe1bcc74e9e3`, cohort, input tensors, endpoint implementation, and seed policy. The seed is $42 +$ `object_idx` and is shared across the matched conditions for each object. Primary and addendum integrity gates pass for 1,200/1,200 and 600/600 condition rows. Fresh C uses the recorded deep/middle/shallow caps of 3.0/3.5/0.8, with requested, applied, and residual scales logged separately. This is a held-out paired comparison under the stated protocol. The historical identity remains explicit: 01549 names v1 weights for its main tables and a separate v2 strong-residual, per-layer-capped instance for Section 4.7/FAC. Fresh C uses a controlled retraining and is not presented as a numerical reproduction of the old Table 4 +0.96 dB estimate.

S9. Historical 01549 Table 9: strict-276 schedule transfer

The 01549 submission reports this schedule-transfer comparison on the 276 objects disjoint from the 24-object probe. It uses the strong-residual, per-layer-capped v2 adapter, not the v1 checkpoint named for 01549 Table 4. The values below are reproduced from the authenticated 01549 submission; they are not pooled with Fresh C or the separate MAIN-STAGE-276 profile-placement analysis. The table therefore documents schedule transfer on its stated adapter instance and remains distinct from Fresh C's matched single-checkpoint comparison.

Table S11: Values reproduced from original 01549 Table 9 for the stated strict-276 retrospective schedule-transfer comparison. The adapter/run differs from Fresh C; row-level runner and input provenance are incomplete. Values are preserved as submitted historical results, not independently verified statistics.

Schedule	FG-SSIM	PSNR	C3–schedule ΔPSNR (95% CI)	C3-favourable objects
Trapezoid	0.356	19.04	+0.258 [+0.207, +0.308]	206/276
Gaussian peak	0.358	18.97	+0.324 [+0.281, +0.367]	232/276
Linear warm-up	0.377	18.97	+0.323 [+0.263, +0.386]	214/276
Cosine bump	0.330	18.39	+0.902 [+0.824, +0.981]	253/276
C3 (TCAS)	0.376	19.29	–	–

S10. Historical 01549 Table 3: extended 24-object development diagnostics

The table below is transcribed from the authenticated 01549 source and moved from the main text to preserve the original 24-object development record. The candidate-level row outputs and selection-time provenance are unavailable for independent reanalysis. Laplacian variance, RGB standard deviation, gradient magnitude, and high-frequency energy are variation diagnostics; this historical table does not establish texture fidelity, a population effect, or an optimal schedule.

Table S12: Values transcribed from original 01549 Table 3 for representative variants in the 24-object development probe. These are historical values, not independently recomputed statistics; variation measures do not establish texture fidelity.

Variant	ΔLap Var	ΔRGB Std	ΔGrad	ΔHF Energy
C3	−0.001	−0.023	+0.142	+0.058
C4	−0.012	−0.032	+0.086	+0.062
A_s2.50	−0.011	−0.044	+0.054	+0.078
A_s2.00	−0.006	−0.040	+0.100	+0.070
A_s1.25	−0.005	−0.032	+0.104	+0.053

S11. First-round FAC controlled re-examination

The 01549 first-round package reports seven trained FAC configurations under the paired 300-object protocol. All values below are retained as originally reported; this is not a new 1008 training run. The 12-object untrained C3-envelope control is separate and is described in the main text.

Table S13: FAC dose-response table reported in the first-round 01549 supplement. FG-PSNR and FG-SSIM are the reported absolute values; ΔFG-PSNR and wins compare each trained configuration with TCAS.

Configuration	Temporal gate	FG-PSNR	FG-SSIM	ΔFG-PSNR	Wins
TCAS baseline (C3)	fixed C3	9.88	0.3681	–	–
LTAG only	joint, unconstrained	7.79	0.3487	−2.08	4/300
LTAG+GSG	joint, unconstrained	7.03	0.3163	−2.84	6/300
Full FAC	joint, unconstrained	6.94	0.3204	−2.94	5/300
Two-stage + GSG/FSC	frozen at C3	7.32	0.3074	−2.56	4/300
Envelope penalty ($\lambda = 1$)	joint, penalized to C3	7.14	0.3021	−2.74	2/300
Fidelity weighting	frozen at C3	7.51	0.3110	−2.37	5/300
Gentle schedule	frozen at C3	8.77	0.4175	−1.10	48/300

The submitted protocol states that only the LTAG/GSG/FSC controllers (roughly 6×10^3 parameters) were trained; the base model, VAE, and GeoTex-Adapter remained frozen. It specifies the enhanced latent denoising objective, AdamW, peak learning rate 10^{-4}, 100 warm-up steps, and 2,000 training steps on a training pool stated as 1,706 objects and disjoint from the 300-object evaluation set. The 12-object untrained warm-start control was reported at ΔFG-PSNR = −0.18 dB, 95% CI $[-0.42, +0.05]$. A current archive audit found a recovered configuration listing 1,200 training objects and no run-bound seed/output manifest. The original result is therefore retained as reported in 01549 with this reproducibility limitation disclosed; no code-release claim is made.

S12. Retrospective texture-complexity boundary

This analysis uses the separate FRESH_CONFIRM_300 cohort ($N = 300$), not Fresh C or Fresh B. LLH and native GFL rows match the frozen object list and share the recorded checkpoint, runner, seed, six unique target views, sampler, and protocol. The paired LLH–GFL mean differences are +1.2877 dB FG-PSNR (95% object-bootstrap CI $[+0.9383, +1.6346]$; 57.7% LLH-favorable objects) and −0.01650 FG-LPIPS ($[-0.01887, -0.01413]$; 73.3% LLH-favorable objects).

The descriptor analysis tests two image-space outcomes against five ground-truth descriptors: foreground Laplacian variance, RGB standard deviation, gradient magnitude, high-frequency energy, and coverage. Spearman correlations use average ranks and paired-object percentile bootstrap intervals (10,000 resamples, seed 20261002). The family contains ten plus-one-corrected bootstrap sign tests and Holm correction across all ten. Nine tests reject at 0.05; the only non-rejection is FG-LPIPS versus coverage (adjusted $p = 0.0508$). The preserved H5 hypothesis excerpt predates one archived outcome bundle, but the main multiplicity lock classifies this family as retrospective after prior outcome exposure; that chronology has not been reconciled. We therefore report all associations as retrospective sensitivity evidence, not as preregistered confirmation.

Ground-truth foreground Laplacian-variance quartiles contain 75 objects each. The FG-PSNR mean LLH–GFL difference declines from +4.140 dB in Q1 (92.0% favor LLH) to +1.127 dB in Q2 (52.0%), −0.087 dB in Q3 (41.3%), and −0.029 dB in Q4 (45.3%). The FG-LPIPS mean changes are −0.03745, −0.01665, −0.00670, and −0.00520 from Q1 to Q4; the favorable-object rates decline from 97.3% to 50.7%. These strata describe heterogeneity in this cohort. They do not establish that texture complexity causes the response, define a threshold, validate an object selector, or measure texture fidelity.

Table S14: Retrospective Spearman associations between LLH–GFL image-space deltas and ground-truth descriptors in FRESH_CONFIRM_300. Correlation intervals are paired-object bootstrap intervals; adjusted p values use the ten-test Holm family with plus-one-corrected bootstrap sign tests.

Outcome delta	Ground-truth descriptor	ρ [95% CI]	Holm p	Decision
FG-PSNR	Laplacian variance	-0.524 $[-0.603, -0.435]$	0.0020	reject
FG-PSNR	RGB standard deviation	-0.380 $[-0.488, -0.266]$	0.0020	reject
FG-PSNR	Gradient magnitude	-0.464 $[-0.548, -0.371]$	0.0020	reject
FG-PSNR	High-frequency energy	-0.691 $[-0.745, -0.626]$	0.0020	reject
FG-PSNR	Foreground coverage	-0.134 $[-0.241, -0.026]$	0.0352	reject
FG-LPIPS	Laplacian variance	$+0.604$ $[+0.511, +0.681]$	0.0020	reject
FG-LPIPS	RGB standard deviation	$+0.379$ $[+0.272, +0.482]$	0.0020	reject
FG-LPIPS	Gradient magnitude	$+0.563$ $[+0.469, +0.643]$	0.0020	reject
FG-LPIPS	High-frequency energy	$+0.677$ $[+0.599, +0.744]$	0.0020	reject
FG-LPIPS	Foreground coverage	-0.118 $[-0.233, +0.0002]$	0.0508	not rejected

S13. Paired color-pathway diagnosis and failure boundaries

The color diagnosis is a selected, failure-enriched set of 26 objects: six earlier figure cases and 20 Fresh C panels. It is not a random sample, a new holdout, or a population estimate. No Adapter, GFL, and LLH were compared on the same object inputs, checkpoint and runner, with six unique target views per object (78 object-condition records). CIEDE2000 and color-difference components were recomputed from hash-identified RGB images and frozen foreground masks. The object-level intervals below are paired percentile bootstrap intervals (20,000 resamples, seed 61008); they are descriptive and no confirmatory multiplicity family was defined for this diagnostic.

Table S15: Selected paired endpoints from the failure-enriched 26-object color diagnosis. Each row is condition A minus B with an object-level 95% paired-bootstrap CI. Lower CIEDE2000 and FG-LPIPS are favorable; higher PSNR and Edge-SSIM are favorable. Local texture statistics describe variation, not fidelity.

Contrast	Endpoint	Mean difference	95% CI
GFL – No Adapter	Foreground CIEDE2000	-19.654	$[-26.370, -13.248]$
GFL – No Adapter	FG-PSNR (dB)	$+6.646$	$[+5.556, +7.635]$
GFL – No Adapter	FG-LPIPS	-0.03251	$[-0.05363, -0.01072]$
GFL – No Adapter	Edge-SSIM	$+0.06510$	$[+0.04853, +0.08202]$
LLH – GFL	Foreground CIEDE2000	$+3.583$	$[+0.462, +6.712]$
LLH – GFL	FG-PSNR (dB)	-0.197	$[-1.562, +1.297]$
LLH – GFL	FG-LPIPS	-0.01294	$[-0.02742, +0.00066]$
LLH – GFL	Edge-SSIM	-0.00080	$[-0.00776, +0.00659]$
LLH – GFL	Foreground gradient magnitude	-0.06772	$[-0.08132, -0.05494]$
LLH – GFL	Foreground high-frequency energy	-0.00887	$[-0.01091, -0.00672]$
LLH – GFL	Foreground color entropy	-0.598	$[-0.703, -0.502]$

GFL lowered reference-relative CIEDE2000 in 24/26 objects, yet the two original Figure 4 cases retained pink or purple shifts under No Adapter, GFL, and LLH. LLH had higher mean CIEDE2000 and lower gradient, high-frequency energy, and color entropy than GFL; these changes do not show improved fidelity. The decoded RGB showed pink before UV projection or baking, and a late denoising snapshot (update 40 of 50) showed it as well. On one selected case, a ground-truth VAE encode–decode isolate gave CIEDE2000 4.086 and $\Delta a^* \approx -1.35$, while GFL gave 22.032 and $+1.977$. This weakens ordinary VAE round-trip error as the sole explanation but does not identify the complete cause.

A feature-relative residual gate was negative: versus C3 it increased mean CIEDE2000 by 1.35 across 26 objects and by 3.18 in the six earlier cases, with no clear visual reduction in pink shift. No further repair search was started because no causal control point was identified. No complete cause or validated repair is established.